

Pablo Silva
Professor Stacey

18.06 - Exam 3

+95 / 100

1. $H = I - 2uv^T$

a. $H^2 = (I - 2uv^T)^2$

$$= \left(\begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} - 2 \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} \right)^2$$

$$= \left(\begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} - 2 \begin{bmatrix} u_1 v_1 & \dots & u_1 v_n \\ \vdots & \ddots & \vdots \\ u_n v_1 & \dots & u_n v_n \end{bmatrix} \right)^2$$

$$= \left(\begin{bmatrix} 1-2u_1 v_1 & -2u_1 v_2 & \dots & -2u_1 v_n \\ \vdots & \vdots & \ddots & \vdots \\ -2u_n v_1 & -2u_n v_2 & \dots & -2u_n v_n \end{bmatrix} \right)^2$$

= Therefore $u_i u_j^T = 0$ for $i \neq j$
 $\therefore H$ must be orthogonal

IMF should have me lost.

2. a. All eigenvalues of A are positive, therefore taking the inverse still results in positive eigenvalues $\Rightarrow$ still positive definite

Since $\det(B) = \begin{bmatrix} \det(A) & \det(C) \\ \det(D) & \det(A) \end{bmatrix}$

Weak

Proof

Here $\det(B) = 0$, then

at least one $\lambda = 0$. Therefore the matrix cannot be positive definite

3. $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

a. $\lambda^2 + 1 = 0 \Rightarrow \begin{bmatrix} 2i & -1 \\ 1 & 2i \end{bmatrix}$

$x = \pm 2i$

$\Rightarrow x_1 = \begin{bmatrix} -i \\ 2 \end{bmatrix}$ $\begin{bmatrix} -2i & 1 \\ 1 & -2i \end{bmatrix}$

$\bar{x}_1 = x_2 = \begin{bmatrix} i \\ 2 \end{bmatrix}$

$v(0) \begin{bmatrix} 2 \\ 0 \end{bmatrix} = i \begin{bmatrix} -i \\ 2 \end{bmatrix} - i \begin{bmatrix} i \\ 2 \end{bmatrix}$

I believe the pset accounts for different combinations.

b. $Hu = (I - 2uv^T)u$

$\Rightarrow Iu - 2uv^T u$
 $= u - 2u = -u$
 $\therefore \lambda = -1$

c. $Hv = (I - 2uv^T)v$

$= v - 2uv^T v = v$
 $\therefore \lambda = 1$

Repeated $n-1$ times and as such H is diagonalizable.

d. As seen already:

$\begin{bmatrix} 1-2u_1 v_1 & & \\ & \ddots & \\ & & 1-2u_n v_n \end{bmatrix} \Rightarrow \begin{bmatrix} 1-2u_1^2 & & \\ & \ddots & \\ & & 1-2u_n^2 \end{bmatrix}$

$\Rightarrow n - \sum_{i=1}^n u_i^2 = n - 2$ with $u_i^2 = 1$
 $\sum \lambda_i = 1 - 1 + n - 1 = n - 1$

b. $Q A Q^T = Q A Q^{-1}$

If Q is orthogonal,

the only way to produce I is by transposing the matrix:

$\begin{bmatrix} q_1 & \dots & q_n \end{bmatrix} \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & \ddots & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \dots & 0 & 1 \end{bmatrix} = I$

Since $Q^T Q = I$, $Q^T = Q^{-1}$

and thus $Q A Q^T = Q A Q^{-1}$

This is the identity/symmetry test.

Oh...?

Now it's actually a thing.

c. $A = U \Lambda U^T$ $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$A^T A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow \Lambda = \sqrt{A^T A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

$\Rightarrow U = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

OK